SHEAF: Secure Heterogeneity-Aware Efficient Aggregation for Federated Learning

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***Abstract*—Federated learning (FL) over communication-constrained edge networks must balance security, model quality, and system cost in the presence of diverse client data. This is a challenge that wasn't addressed holistically. To address this challenge, we propose an aggregation method that quantifies client heterogeneity using the Jensen-Shannon divergence and Shannon entropy, and then fuses this with a peer validation signal to weighted updates. A post-quantum client-server security design is deployed to improve protection against inference and reconstruction attacks. To control bandwidth and computation cost, after experimenting with several methods, weight quantization was applied to ensure that payloads are aligned to resource budgets. On CIFAR-10 under non-IID splits, our aggregation achieves the highest average accuracy in most rounds, reaching 83.59% during rounds 30–59, with a final accuracy of 84.71%. Post-training quantization reduces the transferred model size by approximately 75% while maintaining a roughly 84.9% accuracy.**

***Index Terms*—Federated learning; non-IID data; post-quantum security; cost minimization.**

I. Introduction

The contemporary digital age is deeply data-driven. As the volume and diversity of data continue to expand, there has been a corresponding increase in the regulatory frameworks governing the collection, management, and sharing of data. Federated Learning (FL) addresses these constraints by moving computation to the data, allowing clients to train locally and a server to aggregate their updates into a global model. This approach is vital for sensitive information, as in next-word prediction on mobiles and clinical applications.

A central challenge is that client data are often not independent or identicaly distributed (non-IID), with skew, imbalance, and scale differences that can bias aggregation and reduce overall accuracy. Designs that enhance accuracy under non-IID conditions and those that strengthen security against threats such as membership inference and data reconstruction typically increase computation and communication costs. Our work aims to deliver higher accuracy and faster convergence while controlling costs through performance management techniques that include weight quantization, mixed-precision training, gradient accumulation, selective client participation, and adaptive aggregation scheduling. The resulting contributions are:

- Client updates are weighted using distribution skew metrics, along with peer cross-validation signals, to improve stability and convergence under non-iid data.
- A post quantum ready client server security design is employed and verified through formal analysis using Scyther. Post quantum signatures are incorporated so no client to client trust is required.
- Communication and computation costs are reduced through weight quantization, mixed precision training, and gradient accumulation, applied selectively to meet resource budgets while safeguarding accuracy.

The rest of the paper is organized as follows: section II reviews related work on non-IID mitigation, cost efficiency, and security in FL. Section III presents the SHEAF framework, section IV reports the experimental setup and results including accuracy, convergence, security, and cost analysis, and section V concludes with future directions.

II. Related Work

Early FL aggregates local models by data size as in FedAvg [1], which is sensitive to skew and slows convergence under non-IID data. Accuracy-oriented methods limit client drift or control variance, with FedProx [2] and SCAFFOLD [3] being prominent examples. However, these methods often introduce coordination, auxiliary state, or longer training, and they seldom operate under an explicit resource budget. Bandwidth and model optimization approaches such as periodic averaging with quantization in FedPAQ [4] reduce payloads and rounds, and client selection and scheduling improve wall clock time by favoring high utility participants as in Oort [5]. However, these strands treat cost and accuracy separately and do not focus on explicit non-IID evidence.

Security-focused work strengthens confidentiality and robustness, but rarely accounts for the full cost profiling at the network edge. Secure aggregation secures an honest but curious server [6], while homomorphic encryption offers strong privacy at the expense of inflated message sizes and computations [7]. Post quantum readiness is increasingly relevant as standardized key encapsulation and signatures are adopted [8], yet most designs do not quantify how security choices interact with compression layers or with rounds to target. In contrast, the proposed SHEAF framework treats cost as a first-class constraint alongside accuracy and security by combining heterogeneity-aware and validation-guided weighting with a security design and a set of performance management techniques.

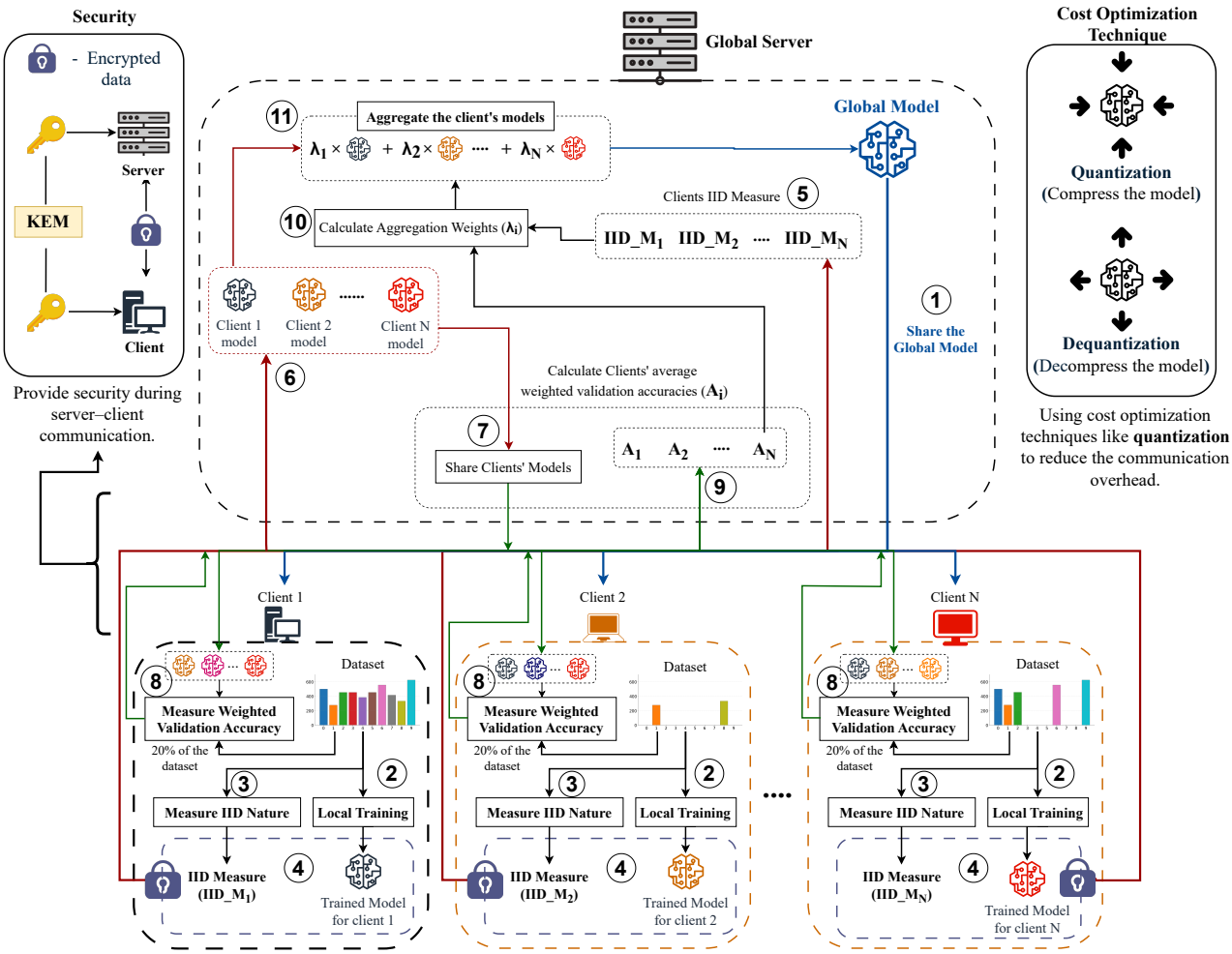


Fig. 1: Proposed FL framework: Secure accuracy-based and non-IID aware aggregation with cost minimization

III. Proposed SHEAF Framework

This study is organized around three components: (i) accuracy enhancement strategies, (ii) integrated security mechanisms for confidentiality and robustness, and (iii) cost minimization with emphasis on cost of communication mitigation. Accordingly, we propose the *Secure Heterogeneity-aware Efficient Aggregation for Federated Learning* (SHEAF) framework, as illustrated in Fig. 1.

Fig. 1 depicts the complete SHEAF workflow. During the initial registration phase, each client and the server perform a *Key Encapsulation Mechanism (KEM)* to establish a shared secret for secure communication. At the beginning of each round, the server *quantizes* and encrypts the *global model* and broadcasts it together with an encrypted session key (*Share the Global Model*, Step 1). Upon reception, each client decrypts the session key, decrypts the received payload, and *dequantizes* (*decompresses*) the model to reconstruct parameters before starting *local training* (Step 2). After training, each client computes the *IID nature* (heterogeneity) of its local dataset to obtain an IID score (Step 3) and produces the trained local update (*Trained Model*, Step 4). The client then *quantizes* the updated model to reduce communication overhead, encrypts the quantized update and IID score using the established secret, digitally signs the transmission for authenticity, and uploads the encrypted update to the server (*Encrypted data*, Step 5). The server verifies signatures, decrypts received updates, and reconstructs client models from their quantized representations (Step 6). To compute *weighted validation accuracy*, the server distributes each client's reconstructed model update to the other clients (*Share Clients' Models*, Step 7). Each client evaluates the received models on a held-out portion of its local data and returns weighted validation accuracy scores (*Measure Weighted Validation Accuracy*, Step 8). The server aggregates these responses to obtain per-client accuracy signals $A_1, \ldots, A_N$ (Step 9) and combines them with the corresponding IID scores to compute aggregation weights λ_i (*Calculate Aggregation Weights*, Step 10). The aggregation weight will be calculated by one of our proposed methods, AB(1-JSD), AB_SEBW, (1-JSD) or SEBW. Finally, the server performs a weighted aggregation of client updates to generate the next global model (*Aggregate the clients' models*, Step 11), which is again quantized, encrypted, and broadcast for the subsequent round until a stopping criterion is met.

A. Accuracy

To mitigate the complications arising from the disparities in client data in an FL round, we propose measuring the non-IID nature of the client data and integrating the acquired information when aggregating the global model. A weight metric is utilized for this purpose.

$$w_g^{t+1} = \sum_{i=1}^{n} \lambda_i w_i^t \tag{1}$$

In Eqn. 1, w_g^{t+1} is the global model at round $t+1$, n is the number of clients, w_i^t is the local model from client i at round t, and λ_i is the explicitly assigned weight for client i. The weights λ_i satisfy $\sum_{i=1}^{n} \lambda_i = 1$. This scales the effect of the client's locally trained model to align with the quality of its inherent data. Several methods are suggested for assigning the λ_i in Eqn. 1 under different requirements.

1) Jensen Shannon Divergence Based Method (1-JSD) : Jensen-Shannon Divergence (JSD) is a symmetric measure of divergence between two probability distributions. To measure the non-IID nature of the client, the JSD of the client's distribution is calculated with respect to an ideal IID distribution.

$$D_{\text{JSD}}(p \parallel q_i) = \frac{1}{2} D_{\text{KL}}\left(p \parallel \frac{p+q}{2}\right) + \frac{1}{2} D_{\text{KL}}\left(q \parallel \frac{p+q}{2}\right) \tag{2}$$

In Eqn. 2, p represents the ideal probability distribution and q represents the client distribution. $D_{\text{KL}}(\cdot \parallel \cdot)$ is the Kullback-Leibler Divergence (KLD), a measure of how one probability distribution diverges from a second reference distribution. JSD is chosen over KLD to ensure symmetry. If a reference IID dataset is available, the ideal distribution p in the JSD-based method can be estimated directly from it. Otherwise, the SEBW method provides an alternative that does not rely on any reference distribution.

Since larger values of D_{JSD} indicate greater dissimilarity from the ideal distribution, to give a higher priority to distributions with an IID nature complement of the divergence is used, as defined in Eq. 3. JSD is computed using base-2 logarithms, which bounds $D_{JSD} \in [0, 1]$.

$$\tilde{\lambda}_i = 1 - D_{\text{JSD}}(p \parallel q_i) \tag{3}$$

The unnormalized weights are then normalized across all participating clients to ensure that the aggregation weights sum to one:

$$\lambda_i = \frac{\tilde{\lambda}_i}{\sum_{k=1}^{n} \tilde{\lambda}_k}, \qquad \sum_{i=1}^{n} \lambda_i = 1. \tag{4}$$

2) Shannon Entropy Based Weighting Method (SEBW) :

$$H(X) = -\sum_{i=1}^{k} p(x_i) \log_2 p(x_i) \tag{5}$$

Using Shannon entropy to quantify the IID nature eliminates the need for a reference dataset. Eqn. 5 denotes the entropy $H(X)$, which quantifies the heterogeneity of the data distribution across classes; and captures the degree of data heterogeneity among clients. Here, $p(x_i)$ is the probability of class x_i in clients' data. To amplify the influence of near-IID data distributions, the Shannon entropy term is squared, a design choice that experimental results show improves convergence stability and final model accuracy by prioritizing clients whose data distributions more closely resemble the IID setting. To incorporate the effect of client data size without allowing it to dominate the aggregation process, we scale the entropy term by $\log_2(\text{client_size})$. The logarithmic form ensures that larger datasets contribute more than smaller ones, while preventing client size from overwhelming the data quality captured by the entropy measure.

Accordingly, the unnormalized entropy-based weight for client i is defined as

$$\tilde{\lambda}_i = H(X)^2 \times \log_2(\text{client_size}) \tag{6}$$

In addition to these statistical methods of measuring the data quality of clients, we combine the accuracy based method of allocating a weight for clients as discussed in [9] to furthur improve the speed of convergence.

3) Combining Accuracy Based Method: After local training, each client uploads its model to the server, which redistributes all client models for peer evaluation among all other clients. Each client uses 20% of its local dataset as a validation set to compute the accuracy of every received model, multiplies each accuracy by the size of the validation set to obtain a weighted validation accuracy, and returns these values to the server. Let the average weighted validation accuracy of client i be denoted by A_i. and the a_{ij} represent the accuracy of the model from client i evaluated on the validation dataset of client j, and let S_j denote the size of client j's validation set. The server computes the average weighted validation accuracy for each client model using Eqn. 7.

$$A_i = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{n} a_{ij} \cdot S_j}{n-1} \tag{7}$$

This metric is subsequently used during the aggregation phase to assign weights to the client models, guiding the generation of the new global model. Combining a statistical measure with the accuracy-based value, we introduce Accuracy-Based with Jensen–Shannon Divergence (AB(1-JSD)) and Accuracy-Based with Shannon Entropy-Based Weighting (AB_SEBW).

$$\tilde{\lambda}_i = A_i \cdot Agg_IM_i \tag{8}$$

The combined aggregation weight $\tilde{\lambda}_i$ for each client is calculated by multiplying the average weighted validation accuracy A_i with the statistical measure aggregation weight Agg_IM_i, as defined in Eqn. 8. These methods aim to enhance both model accuracy and convergence speed, assuming sufficient computational resources are available. If not, the accuracy-based component can be omitted to simplify. The choice of statistical measure depends on the availability of a reference IID dataset. If available, the (1-JSD) method can be applied; otherwise, the SEBW method is recommended.

B. Post-quantum Light-weight Security

To mitigate the overhead of client-to-client communication, we adopt a Client-Server architecture protected by a Post-Quantum Cryptography (PQC) protocol Fig 2) designed under a Dolev-Yao threat model. We consider a threat model in

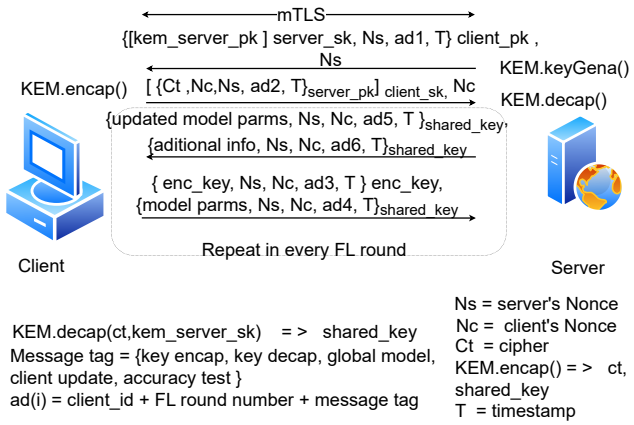


Fig. 2: Proposed security protocol for SHEAF

which clients may behave maliciously, while the server is trusted and the communication network is adversarial. The primary focus of this cryptographic protocol is to secure the data exchange over the untrusted network transport layer, explicitly addressing external threats and man-in-the-middle attacks. This addresses active network adversaries attempting eavesdropping or tampering; PQC is essential as trained models must be secured against future quantum decryption threats that render RSA/ECC vulnerable. For quantum-resistant key exchange, Rudraksh KEM poly 64 is selected for resource-constrained devices (IoT) [10], while ML-KEM-768 (Kyber) is used for general-purpose nodes. For data encryption, we employ a dynamic selection strategy: Ascon-80pq for lightweight AEAD on constrained devices, and AES-GCM-256 where hardware acceleration is available. Authenticated integrity is enforced via Falcon-512 signatures within an mTLS handshake, establishing strict mutual authentication. The protocol incorporates unique nonces (Nc, Ns) and verifiable timestamps (T) within all communication messages, providing message freshness and ensuring resilience against replay attacks by preventing the reuse of intercepted data. Each client then performs a key exchange with the server to create a unique shared secret key ($shared_key$); local model parameter updates are encrypted with $shared_key$, the server decrypts and aggregates them, and a new global model is produced.

To ensure scalability during model distribution phases (both peer validation and global updates), we implement a key-wrapping optimization: the server encrypts the heavy global model only once using a fresh symmetric key (enc_key), and subsequently encrypts only the lightweight enc_key individually for each client using their shared secrets. This reduces the server's encryption complexity from linear $O(N)$ to constant time $O(1)$. The protocol's security properties, including Secrecy and Authentication (Aliveness, Weak Agreement), were formally verified using the Scyther tool. Finally, while the PQC layer secures the transport channel, robustness against insider model poisoning is achieved through the proposed AB(1-JSD) aggregation mechanism, which algorithmically filters malicious updates by assigning them negligible weights based on heterogeneity and validation metrics.

C. Cost minimization

Cost minimization in FL is closely tied to both communication and computation efficiency. To minimize these costs of the proposed FL framework while preserving the high accuracy achieved through SHEAF framework, we tested several cost minimization techniques focused on alleviating the said overheads. These techniques include weight quantization, model pruning, mixed-precision training, and gradient accumulation, applied selectively to minimize resource consumption without compromising model effectiveness, while accelerating training. This makes SHEAF more suitable for resource constrained environments.

Weight quantization is employed to compress model parameters from full-precision (32-bit floating-point) to lower-precision formats (8-bit integers), significantly reducing model size and communication costs. Mixed-precision training leverages hardware capabilities (Tensor Cores in GPUs) by using lower-precision formats, FP16 for most computations while retaining FP32 for critical operations to avoid numerical instability, and gradient accumulation aggregates gradients over multiple mini-batches before updating parameters, accelerating training by reducing memory usage and enabling larger effective batch sizes. However, mixed-precision and gradient accumulation can degrade accuracy due to precision-related instabilities in heterogeneous data settings. To balance these trade-offs, weight quantization is retained as the sole cost minimization technique in the final framework, reducing communication and memory overhead while maintaining global accuracy within $\pm 1\%$ of the SHEAF baseline. Quantization is applied post-local training on clients and reversed during server-side aggregation.

Let W_{server} denote the global model weights broadcast by the server at the beginning of a round, and let W_{client_i} denote the local model update (weights) produced by the i-th selected client and uploaded to the server at the end of the round. Let $S(W)$ be the size (in bytes) of uncompressed weights and $S(Q(W))$ be the size after quantization/compression. For n selected clients, the per-round communication cost is given in Eqn. 9, and the corresponding quantized (compressed) cost is given in Eqn. 10.

$$C_{\text{total}} = n \cdot S(W_{\text{server}}) + n \cdot \sum_{i=1}^{n} S(W_{\text{client}_i}) \tag{9}$$

$$C_{\text{compressed}} = n \cdot S(Q(W_{\text{server}})) + n \cdot \sum_{i=1}^{n} S(Q(W_{\text{client}_i})) \tag{10}$$

Further, the proposed secure FL protocol was integrated and evaluated in terms of communication overhead. Although the use of PQC primitives introduces additional computational cost for key exchange and authentication, the protocol is optimized to minimize server-side overhead during model distribution. Specifically, the global model parameters are encrypted only once at the server, and the corresponding encryption key is subsequently wrapped and individually encrypted using the established shared key of each client. This design avoids repeated encryption of the full model for every client and significantly reduces computational load. In addition,

applying weight quantization substantially reduces the size of the model parameters, which lowers both the encryption cost and the required transmission bandwidth. As a result, the reduction in size of the model parameters compensates for the cryptographic overhead, leading to an overall improvement in communication efficiency in the secure setting.

The proposed SHEAF method enhances federated learning by integrating heterogeneity-aware weighted aggregation using AB(1-JSD), AB_SEBW, SEBW or 1-JSD for accuracy, quantization for increasing communication and computation cost efficiency, and a robust security protocol for improved data protection and efficiency.

IV. Experimental Results

A. Experimental Setup

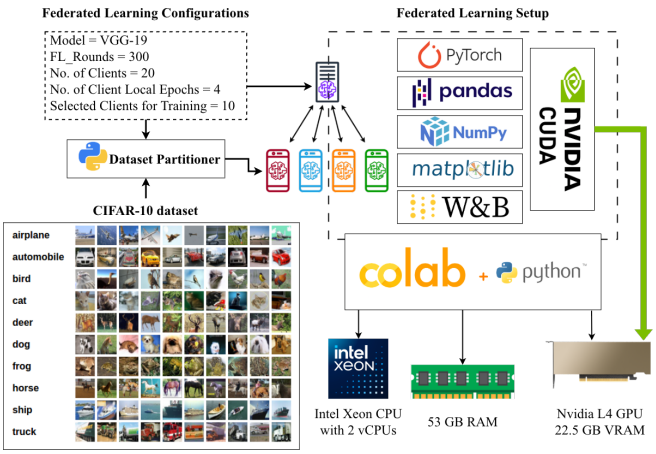


Fig. 3: Experimental setup

The experiments used the CIFAR-10 and 5G-NIDD [11] datasets. CIFAR-10 consists of 60,000 color images across 10 categories (6,000 per class), with 50,000 training and 10,000 testing images of size 32×32 pixels. The 5G-NIDD dataset contains real and simulated 5G network traffic for intrusion detection, covering multiple benign and attack types. The setup involved 20 clients, each training a VGG-19 model for CIFAR-10 and a simple PyTorch-based neural network for 5G-NIDD to predict intrusions, trained for 4 local epochs per communication round over 300 rounds (60 for 5G-NIDD). The partitioned client datasets exhibit substantial heterogeneity, with average data skew scores of 2.2417 for CIFAR-10 and 1.3438 for 5G-NIDD, computed over 20 trials using the data skew metric defined in [12].

The implementation, developed in Python using NumPy and PyTorch, was executed on Google Colab with an NVIDIA L4 Tensor Core GPU (22.5 GB VRAM) and an Intel Xeon CPU (2 vCPUs, 53 GB RAM). The experimental setup for the CIFAR-10 dataset is shown in Fig. 3. The implementation is publicly available [1].

B. Client Dataset Distribution and Measuring IID Nature

The CIFAR-10 dataset is partitioned among 20 clients using a custom Dataset Partitioner that allocates image classes to

[1]https://github.com/FYP-G-16-UOR-FOE/FL-Implimentation

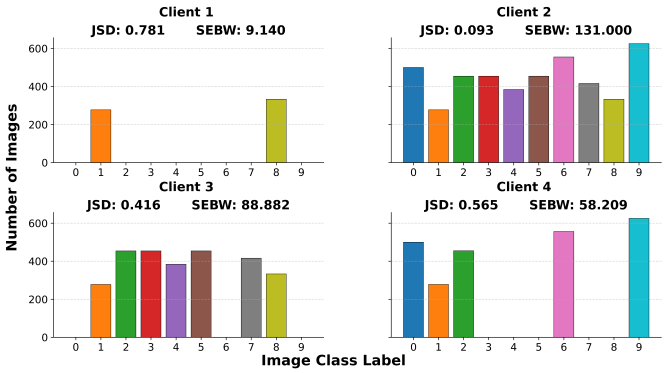
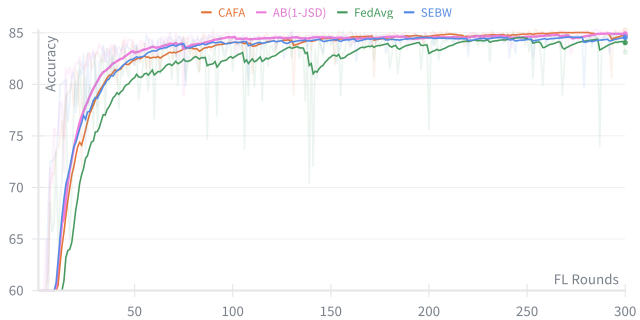


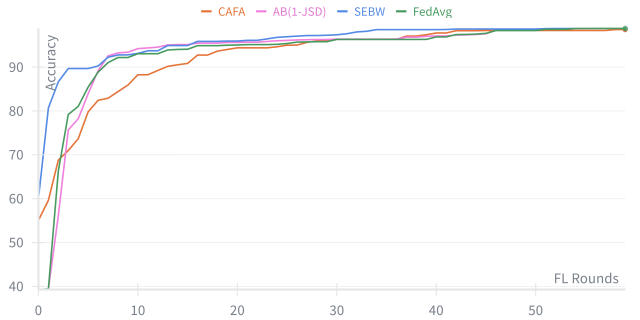
Fig. 4: Client dataset distribution and IID nature measures (JSD and SEBW) for four selected clients.

simulate realistic data heterogeneity. As illustrated in Fig. 4, the class distributions of four selected clients demonstrate varying degrees of IID and non-IID characteristics. Client 1 contains only two image classes, indicating a highly non-IID nature, with a JSD of 0.781 and an SEBW of 9.140. In contrast, Client 2 includes all ten image classes, representing a nearly IID distribution (JSD = 0.093, SEBW = 131.000). Client 3 exhibits moderate diversity (JSD = 0.416, SEBW = 88.882), while Client 4 lies between non-IID and IID (JSD = 0.565, SEBW = 58.209). These results indicate that a lower JSD value and higher SEBW value correspond to a distribution closer to IID, whereas higher JSD and lower SEBW values signify a more non-IID dataset.

C. Analysing the accuracy of SHEAF



(a) CIFAR-10 test accuracy over 300 rounds.



(b) 5G-NIDD test accuracy over 60 rounds.

Fig. 5: Comparison of accuracy between the proposed SHEAF methodS (AB(1-JSD),SEBW) and FedAvg, CAFA.

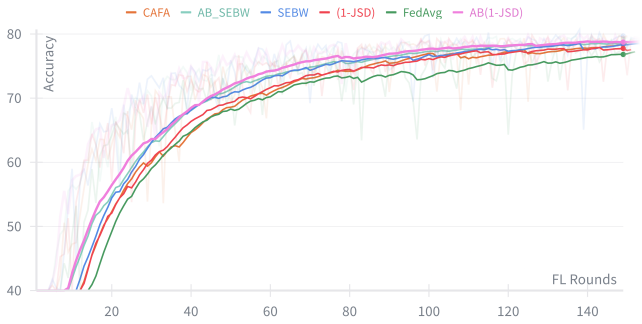


Fig. 6: CIFAR-10 test accuracy: 100 clients, 150 rounds.

As shown in Table I and Fig. 5, the four proposed methods, namely 1-JSD, SEBW, AB_SEBW, and AB(1-JSD), consistently outperform the baseline FedAvg as well as the recently introduced CAFA method. To improve the interpretability of the accuracy curves, the Time Weighted Exponential Moving Average (TWEMA) algorithm is applied to smooth the results.

TABLE I: Average accuracy across training rounds for CIFAR-10 and 5G-NIDD datasets.

Dataset	Method	0–29	30–59	60–99	100–199
CIFAR-10	**FedAvg**	66.06%	81.00%	82.49%	83.11%
	CAFA	70.22%	82.81%	83.74%	84.39%
	1-JSD	70.24%	82.30%	83.65%	84.14%
	SEBW	71.71%	83.00%	83.98%	84.30%
	AB_SEBW	**72.07%**	83.36%	84.05%	84.41%
	AB(1-JSD)	72.01%	**83.59%**	**84.23%**	**84.50%**
5G-NIDD	**FedAvg**	88.25%	97.29%	—	
	CAFA	86.65%	97.28%	—	
	1-JSD	90.09%	98.36%	—	
	SEBW	**92.43%**	**98.41%**	—	
	AB_SEBW	90.58%	97.93%	—	
	AB(1-JSD)	88.36%	97.54%	—	

For CIFAR-10, AB(1-JSD) achieves the highest average accuracy across most communication rounds, particularly between rounds 30–199, indicating that the model is less affected by fluctuations caused by the non-IID nature of the data. Similarly, on the 5G-NIDD dataset, SEBW achieves the best overall accuracy of 98.41%, while AB(1-JSD) and AB_SEBW also demonstrate competitive stability and convergence during the early rounds (0–59).

In our initial experiments, the CIFAR-10 dataset was partitioned across 20 clients. To further enhance data heterogeneity and evaluate robustness under a more distributed setting, we conducted an additional set of experiments by increasing the number of clients to 100. As depicted by the Figure 6, it performs similar to the previous results in Table I.

It is worth noting that AB(1-JSD) and AB_SEBW naturally introduce higher communication overhead compared to FedAvg, 1-JSD, and SEBW, as a trade-off for improved accuracy. This trade-off is reflected in a 23.8% increase in total training time for the CIFAR-10 dataset under identical experimental conditions. Although AB(1-JSD) achieves better overall performance, its accuracy improvement over SEBW and 1-JSD is relatively small for certain dataset distributions. Thus, from a cost minimization perspective, SEBW and 1-JSD can be considered more favorable due to their lower computational and communication overhead.

Therefore, within the SHEAF framework, the choice of heterogeneity measurement mechanism should be guided by the available computational resources and the presence of reference data. AB(1-JSD) is recommended when a reference dataset is available and sufficient computational capacity can be afforded, whereas AB-SEBW is appropriate in scenarios where no reference data exist and computational resources are not constrained. In resource-limited environments without reference data, SEBW is the preferred option. Conversely, when lightweight computation is required and a reference dataset is available, 1-JSD offers an effective alternative.

D. Analysis of PQC security

Implemented the Rudraksh (poly 64) and used liboqs for ML-KEM-768, Falcon-512, and ML-DSA-44, with the Python `cryptography` library providing ECDH, ECDSA, and AES-GCM-256, and all tests executed in Google Colab on a T4 GPU with 12.7 GB of system RAM.

Results show ML-KEM-768 was significantly faster than both Rudraksh and ECDH: encapsulation and decapsulation times were 0.00018s and 0.00008s for ML-KEM-768, versus 0.00100s and 0.00025s for ECDH, and 0.00300s and 0.00330s for Rudraksh. While Rudraksh is designed for constrained devices, ML-KEM is more performant for devices with more resources; as both offer post-quantum security, the selection depends on target device capabilities. In symmetric encryption tests, AES-GCM-256 was faster (0.0699s / 0.0669s) than Ascon80pq (16.619s / 16.762s) and Ascon128a (16.829s / 16.653s), making AES-GCM-256 suitable where hardware support is available and classical security is sufficient, while Ascon80pq is suited for constrained IoT cases that require robustness against post-quantum threats. Communication overhead was negligible, with encrypted model size increasing only from 82,345,811 to 82,345,827 bytes across algorithms.

TABLE II: Digital Signature Algorithm Comparison (Post-Quantum(PQ) and Conventional(C))

Algorithm	Class	Public Key (bytes)	Signature (bytes)	Signing Time (s)	Verify Time (s)
Falcon-512	PQ	897	690	0.000550	0.00019
ECDSA	C	64	256	0.000250	0.00024
ML-DSA-44	PQ	1312	2420	0.000233	0.00022

SHEAF security protocol uses a digital signing algorithm for mutual authentication and data integrity, prioritizing minimal computational and communication overhead for the FL process. In the selected PQC algorithms, a minimal signature size was more critical than signing speed, as authentication is infrequent. Falcon-512 was selected, based on the observations of Table II. It is a quantum-resistant algorithm with a minimal signature size, yielding reduced communication overhead.

Furthermore, we formally verified the security protocol using the Scyther tool. The analysis tested claims for authentication (Alive, Niagree, Nisynch) and confidentiality (Secrecy of shared Key, enc Key, model, and updated model parameters). The results verified that all claims hold, proving that the protocol is robust against active adversaries such as tampering, impersonation, and outsider attackers, including eavesdropping. The full Scyther protocol specification and verification results are available at our GitHub repository [1].

E. Cost Minimization Analysis

To minimize cost in SHEAF, we applied the setup in Section IV for 150 FL rounds, focusing on the metrics listed in Table III, which summarizes the cost-minimization techniques.

TABLE III: Performance of cost minimization techniques with AB(1-JSD) without the security protocol

Technique	Total FL Training Time Improvement	Total Client Training Time Improvement	Total Size -Quantized (MB)	Transferred Model Size (MB)	Final Global Accuracy	Avg GPU Mem (MB)
AB(1-JSD)	–	–	–	78.45	84.9%	1580
AB(1-JSD) + Quantization	3.606%	3.606%	58.87	19.62	84.87%	1854
AB(1-JSD) + Model Pruning	–	–	23.51	54.94	83.21%	2200
Mixed Precision + Gradient Accumulation	31.074%	41.48%	–	78.45	63.06%	1530

Incorporating quantization into the baseline refines communication efficiency. This variant reduces the transferred model size by 75%. The reduction in training time is modest, since quantization mainly affects inference and communication overheads and does not substantially reduce the cost of local training computations. Critically, accuracy is preserved at 84.87%, demonstrating that quantization maintains model fidelity in non-IID settings without introducing significant precision loss.

Mixed precision with FP16 and FP32 accumulators plus gradient accumulation over four mini-batches cuts total training time by about 31 percent and improves data transmission rate, with a small increase in GPU memory, but accuracy drops to 63.06 percent, which is about 24.4 percent below the baseline. Overall, the SHEAF framework with quantization is the most practical choice, offering more than 3% faster training, approximately 75% smaller model transfers, and accuracy above 84%. Future work can explore adaptive precision scaling to keep the speed benefits while recovering accuracy. To evaluate the impact of embedding security and cost minimization techniques into SHEAF, overall findings are depicted in Table IV.

TABLE IV: Integrating Quantization for SHEAF

Method	Average Accuracy	Average FL Round Time	Model Ecryption Time	Model Size
AB(1-JSD)	83.19%	19.59 s	-	78.45 MB
AB(1-JSD)+Q	83.16%	19.42 s	-	19.62 MB
AB(1-JSD)+Sec	82.81%	61.33 s	0.0672 s	78.53 MB
AB(1-JSD)+Sec+Q	82.79%	54.58 s	0.0097 s	19.64 MB

Running the AB(1-JSD) for the 300 FL rounds, it is apparent that integrating neither the security mechanisms nor the quantization strategies significantly impacts the global accuracy of the framework. However, it is noteworthy that total time per federated round significantly increases with the addition of the security mechanisms by 3.13 times and the quantization algorithm successfully reduces the average FL round time by 61.33 s to 54.58 s, showing an 11% improvement. This improvement can be credited to the model size reduction by quantization, subsequently reducing the encryption time.

V. Conclusion

Prior work on non-IID federated learning often incurs high computational and communication costs for modest accuracy gains, leaving security gaps. We present a unified approach that measures client heterogeneity with Jensen–Shannon divergence and Shannon entropy, blends an accuracy signal into aggregation, strengthens training with a post-quantum cryptographic protocol, and lowers cost with quantization. In experiments, we observe higher accuracy, faster convergence, stronger security, and a modest reduction in training rounds. Future work will investigate the incorporation of model pruning and mixed-precision training to further reduce computational costs. In addition, we plan to evaluate the impact of performing peer validation using only a subset of clients, with the goal of improving the scalability of the proposed approach.

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